

# Dubai DB1 transaction pipeline column documentation

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Source file:

D:\Dubai\Dubai\_Updated\_data\Raw\_Data\Transactions.csv

This pipeline creates the DB1 transaction output by:

1. Reading raw Dubai transaction records.
2. Deriving date and property classification columns.
3. Renaming raw columns to the expected DB1 column names.
4. Creating missing DB1 schema columns with default values.
5. Reordering selected important columns at the front of the output.

## Derived columns and applied logic

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instance\_date:

Source: raw instance\_date.

Logic: converted from string to pandas datetime using format '%d-%m-%Y'.

Invalid dates are converted to NaT because errors='coerce' is used.

year:

Source: instance\_date after datetime conversion.

Logic: extracts the calendar year using instance\_date.dt.year.

Stored as pandas nullable integer Int64 so missing dates remain <NA> instead of becoming float values.

quarter\_num:

Source: instance\_date after datetime conversion.

Logic: extracts quarter number 1, 2, 3, or 4 using instance\_date.dt.quarter.

Stored as nullable integer Int64.

Note: this is an intermediate helper column and is not included in the rename mapping.

quarter:

Source: quarter\_num and year.

Logic: concatenates values into 'Q<quarter\_num>-<year>', for example 'Q4-2006'. If instance\_date is invalid or missing, quarter is set to None to avoid values like 'Q<NA>-<NA>'.

property\_type:

Source: property\_sub\_type\_en.

Logic:

- Flat if property\_sub\_type\_en is one of:  
Flat, Villa, Stacked Townhouses, Unit.
- Shop if property\_sub\_type\_en is one of:  
Shop, Store, Show Rooms.
- Office if property\_sub\_type\_en is one of:  
Office, Clinic, Workshop, Warehouse.
- Others for every other value, including the values listed in the others set and any unexpected or missing subtype.

## Renamed DB1 columns

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These columns are created by renaming raw source columns. No calculation is

applied except where the source column itself was already derived earlier.

|                              |                                         |
|------------------------------|-----------------------------------------|
| project_id                   | <- proj_id                              |
| index                        | <- project_number                       |
| project_name                 | <- project_name_en                      |
| village_name_marathi         | <- village_mr                           |
| location_id                  | <- loc_id                               |
| location_name                | <- area_name_en                         |
| location_name_ar             | <- area_name_ar                         |
| village_name                 | <- igr_village                          |
| year                         | <- year, derived from instance_date     |
| quarter                      | <- quarter, derived from instance_date  |
| city_id                      | <- city_id                              |
| city_name                    | <- city, later overwritten as 'Dubai'   |
| document_number              | <- transaction_id and document_no       |
| sub_registrar_office_code    | <- sro_code                             |
| sub_registrar_office_name    | <- sro_name                             |
| transaction_type             | <- procedure_name_en                    |
| agreement_price              | <- actual_worth                         |
| guideline_value              | <- market_value                         |
| property_description         | <- property_description                 |
| transaction_date             | <- instance_date                        |
| floor_number                 | <- floor_no                             |
| unit_number                  | <- unit_no                              |
| net_carpet_area_sq_m         | <- procedure_area                       |
| balcony_sq_m                 | <- balcony_sqmt                         |
| terrace_sq_m                 | <- terrace_sqmt                         |
| seller_name                  | <- seller_name                          |
| buyer_name                   | <- purchaser_name                       |
| transaction_category         | <- property_category and trans_group_en |
| internal_document_number     | <- internaldocumentnumber               |
| micr_number                  | <- micrno                               |
| bank_type                    | <- bank_type                            |
| party_code                   | <- party_code                           |
| date_of_agreement_execution  | <- dateofexecution                      |
| stamp_duty_paid              | <- stampdutypaid                        |
| registration_fee             | <- registrationfees                     |
| property_type_raw            | <- property_sub_type_en                 |
| unit_configuration           | <- rooms_en                             |
| unit_configuration_ar        | <- rooms_ar                             |
| buyer_pincode                | <- buyer_pincode                        |
| buyer_locality               | <- locality_of_buyer                    |
| buyer_district               | <- district                             |
| buyer_state                  | <- statename                            |
| tower_name                   | <- building_name_en                     |
| tower_name_ar                | <- building_name_ar                     |
| gross_carpet_area_sq_ft      | <- gross_carpet_sqft                    |
| price_per_sq_ft_gross_carpet | <- rate_on_gca_sqft                     |
| is_duplicate                 | <- is_duplicate                         |
| sale_type                    | <- primary_sale_or_secondary_sale       |
| project_type                 | <- project_type                         |

Important note about duplicate target names:

document\_number appears twice in the mapping:

transaction\_id -> document\_number  
document\_no -> document\_number  
transaction\_category also appears twice:  
property\_category -> transaction\_category  
trans\_group\_en -> transaction\_category  
pandas rename will allow duplicate column names if both source columns are present. The pipeline does not currently choose one source over the other.

### Columns created with null/default values

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These expected DB1 columns do not come from the raw source in this script. They are first created as None if missing:

project\_latitude  
project\_longitude  
location\_latitude  
location\_longitude  
is\_ilm\_processed  
is\_manual\_processed  
country\_name  
state\_name  
micro\_market  
sub\_locality  
pincode  
parking\_count  
facing\_direction  
view\_type  
furnishing\_status  
condition\_status  
source\_accessibility  
source\_accessibility\_way  
sourcing\_cost  
sourcing\_time  
data\_type  
data\_source

After creation, these columns are overwritten with fixed values:

city\_name = 'Dubai'  
state\_name = 'Dubai'  
country\_name = 'United Arab Emirates'  
is\_ilm\_processed = 'No'  
is\_manual\_processed = 'No'  
source\_accessibility\_way = 'Download'  
source\_accessibility = 'Easy'  
data\_type = 'Registered Document'  
data\_source = 'DLD'

Columns that remain None unless populated elsewhere:

project\_latitude, project\_longitude, location\_latitude,  
location\_longitude, micro\_market, sub\_locality, pincode, parking\_count,  
facing\_direction, view\_type, furnishing\_status, condition\_status,  
sourcing\_cost, sourcing\_time.

### Output order logic

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The pipeline moves the following columns to the front if they exist:

index, project\_name, project\_name\_ar, quarter, year, location\_name,  
location\_name\_ar, city\_name, property\_type, project\_latitude,  
project\_longitude, location\_latitude, location\_longitude.

All other columns keep their existing order after these priority columns.